SUPPLY CHAIN & PROCUREMENT PERFORMANCE ANALYSIS

1.Project Overview and Objective:

- The Supply Chain & Procurement Performance Analysis project focuses on analysing purchase orders, supplier performance, warehouse operations, order fulfilment, and delivery timelines. The dataset contains 18,000 purchase-order records and 13 columns, including order details, supplier information, warehouse locations, ordered quantities, received quantities, and order values.
- This project uses Microsoft Excel and Power BI features such as data cleaning, Pivot Tables, Power Query, DAX measures, charts, and interactive dashboards to evaluate procurement performance, identify delivery delays, compare supplier performance, and track order fulfilment. The analysis aims to transform raw procurement data into meaningful insights that support better business decisions.

2.Data Sources:

- **Source Description and Timeline:** Inventory & Supply Chain Dataset for Analysis
- **Domain:** Supply Chain / Procurement Analytics.

3.Problem Statement:

- Businesses manage numerous purchase orders across different suppliers and warehouse locations, making it difficult to track procurement spending, order fulfilment, and delivery performance.
- The project aims to analyse procurement data to identify delivery delays, compare supplier performance, monitor ordered versus received quantities, and understand spending trends.
- The analysis helps identify supply chain inefficiencies and supports better decision-making to improve overall operational efficiency.

4. Attribute (Column / Features) Details

The cleaned dataset contains 18,000 purchase-order records and 19 attributes. The following table describes the fields used in the analysis.

Attribute Name	Data Type	Description
purchase_order_id	Text	Unique identifier assigned to each purchase order.
ordered_at	Date	Date on which the purchase order was placed.
expected_at	Date	Expected delivery date for the purchase order.

received_at	Date	Actual date on which the order was received; blank when no receipt date is recorded.
Month	Text	Month label associated with the order date.
Delivery delay days	Whole Number	Difference between the actual and expected delivery dates, expressed in days.
Delivery performance	Text	Delivery timing category: Early, On Time, or Delayed.
Received_status	Text	Receipt completion category, such as Fully Received, Partially Received, or not received.
Status	Text	Purchase-order status, such as Received, Partial, Open, or Cancelled.
supplier_id	Text	Unique identifier for the supplier.
country_code	Text	Short country code associated with the order.
country _names	Text	Country name associated with the order.
warehouse_id	Text	Identifier of the warehouse handling the order.
warehouse_region	Text	Region associated with the warehouse.
line_count	Whole Number	Number of line items recorded for the purchase order.
quantity_ordered	Whole Number	Total quantity requested in the purchase order.
quantity_received	Whole Number	Quantity received against the purchase order.
quantity not received	Whole Number	Quantity ordered but not yet received.
order_value	Decimal Number	Total value recorded for the purchase order.

5. Tools & Technologies

Microsoft Excel: Used for reviewing, cleaning, and preparing the dataset, including handling missing values and checking field consistency.

Power BI Desktop: Used to import the cleaned dataset, perform data modelling, create DAX measures, and build interactive visualizations and dashboards.

Power Query: Used within Power BI for data transformation and preparation before loading the data into the report.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed

Data Cleaning & Transformation:

- Checked and removed duplicate records.
- Identified missing values in the received_at column.
- Standardized date and numerical data formats.
- Created calculated fields such as delivery delay and quantity not received.

Filtering & Sorting:

- Filtered and organized records based on order status, supplier, warehouse, and delivery performance.
- Sorted date and month fields chronologically.
- Identified delayed and incomplete orders.

Pivot Tables:

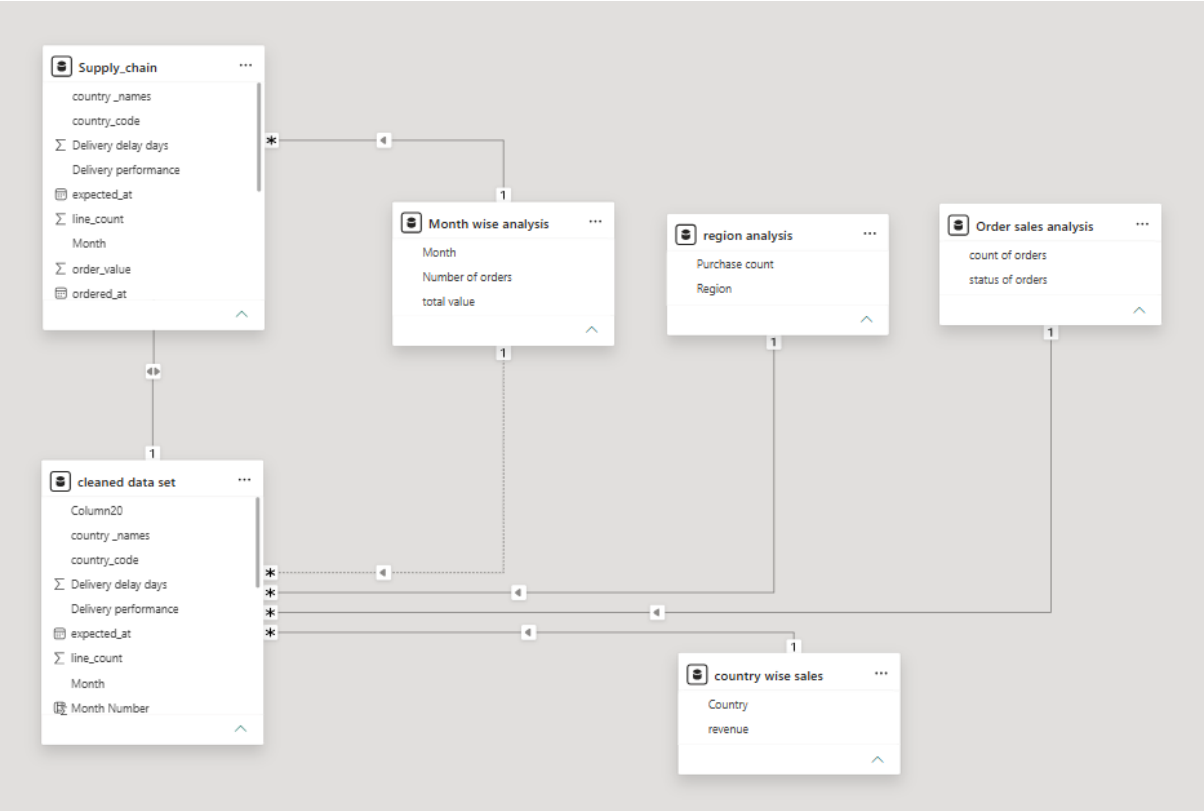
- Summarized monthly order value and order status.
- Analyzed supplier and warehouse performance.
- Compared ordered and received quantities.
- Generated initial insights into procurement and delivery trends.

7. Data Modelling and DAX (Power BI)

Data Model

- Established relationships between the cleaned data set table and the other analysis tables.
- Defined table relationships using one-to-many and many-to-one cardinality, as applicable.
- Used date, country, region, and order-related tables to support data analysis.
- Created a separate Measure table to organize DAX measures.
- Configured relationships and filter directions to support data visualization and reporting.

Screenshot of Data Modelling:



Calculated Columns & DAX Measures

DAX Measures Created:

- Total Orders – Calculates the total number of purchase orders.
- Total Order Value – Calculates the total value of all orders.
- Total Quantity Ordered – Calculates the total quantity ordered.
- Total Quantity Received – Calculates the total quantity received.
- Total Suppliers – Calculates the number of suppliers.
- Delayed Orders – Calculates the number of orders that were delayed.

Calculated Columns Created:

- Delivery Delay Days – Calculates the number of days between the expected delivery date and the actual received date.
- Delivery Performance – Categorizes orders based on their delivery performance.

8. Analysis and Visualizations (Power BI)

Dashboard Features

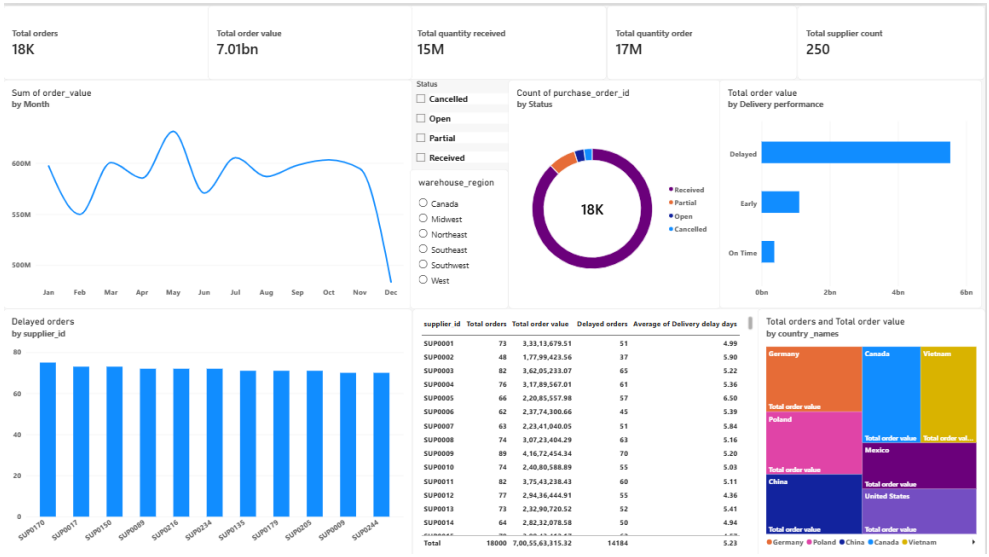
- Multiple Visualizations Based on the Problem Statement:

- KPI cards displaying total orders, total order value, total quantity received, total quantity ordered, and total supplier count.
- Line chart showing monthly procurement value trends.
- Donut chart showing the distribution of purchase orders by status.
- Bar chart comparing total order value by delivery performance.
- Bar chart displaying delayed orders by supplier.
- Table summarizing supplier-wise total orders, order value, delayed orders, and average delivery delay.
- Treemap comparing total orders and order value across countries.

- Interactive Report Layout:

- Added slicers for order status and warehouse region.
- Enabled filtering of visuals based on selected categories.
- Used interactive visual selections to explore supplier, country, and delivery performance details.
- Added clear titles to each visual for easy understanding.
- Displayed data labels and category names wherever applicable.

Organized visuals to provide an overview of order volume, procurement value, delivery performance, supplier delays, and country-wise analysis.Designed the dashboard to support quick monitoring and interactive analysis.



9. Insights & Conclusions

Key Findings

- The dataset contains 18,000 purchase orders from 250 suppliers.
- The total order value is approximately 7.01 billion.
- The total quantity ordered is approximately 17 million, while the quantity received is approximately 15 million.
- The dashboard indicates that delayed orders account for a significant portion of the total order value.
- Supplier-wise analysis shows differences in the number of delayed orders and average delivery delay.
- Country-wise analysis highlights variations in procurement value across different countries.
- The order status analysis shows the distribution of orders across Received, Partial, Open, and Cancelled categories.

Analysis Insights

● Descriptive Analysis

- Summarized total orders, order value, quantities, and supplier count.
- Analyzed monthly procurement value trends.
- Examined order status distribution and country-wise procurement value.
- Compared supplier delivery performance and delayed orders.

● Diagnostic Analysis

- Identified suppliers with higher numbers of delayed orders.
- Compared average delivery delay across suppliers.
- Examined delivery performance categories to identify areas requiring further investigation.
- Analyzed differences between ordered and received quantities.

● Predictive Analysis

- Historical monthly procurement trends can be used as a baseline for future procurement planning.
- Supplier delivery history may help identify suppliers who could be at risk of future delays.
- The current dashboard provides historical analysis.

● Prescriptive Analysis

- Review suppliers with high delayed-order counts and investigate the reasons for delays.
- Monitor the difference between ordered and received quantities to improve fulfillment tracking.

- Prioritize follow-up on open and partially received orders.
- Use monthly procurement trends and supplier performance metrics to support purchasing and delivery planning.

10. Conclusions

The integration of Excel and Power BI proved effective for end-to-end supply chain data analysis, from data cleaning and preprocessing to interactive visual reporting. The project provided insights into procurement value, order status, supplier performance, delivery delays, and quantity fulfillment.

The consolidated Power BI dashboard enables users to monitor key performance indicators, identify delayed orders, compare supplier performance, and analyze procurement trends across countries and warehouse regions. These insights can support better procurement planning, supplier monitoring, and delivery performance improvement.